Power Series

How does a scientific calculator work? Whether it stores all those values?

Power series

- Power series centered at c : If x is a variable, then an infinite series of the form
$$\sum_{n=0}^{\infty} a_n(x-c)^n = a_0 + a_1(x-c) + a_2(x-c)^2 + ...$$
is called a power series centered at c, where $a_n \in \mathbb{R}$ represents the coefficient of the nth term and $c \in \mathbb{R}$ is a constant.
- Power series centered at 0 : An infinite series of the form
$$\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + ...$$
is called a power series centered at 0.
- **Note:** $\sum_{n=0}^{\infty} a_n(x-c)^n = a_0 + a_1(x-c) + a_2(x-c)^2 + ...$ always converges for $x = c$ and the sum of the series is a_0.

Power series

Theorem

For a power series $\sum_{n=0}^{\infty} a_n(x-c)^n$ exactly one of the following three cases is true:

- **Case 1:** The series converges only for $x = c$.
- **Case 2:** The series converges for all x.
- **Case 3:** There exists a positive real number R such that the series converges absolutely for all real x satisfying $|x-c| < R$ and diverges for all x satisfying $|x-c| > R$.

Radius of Convergence

- The radius of convergence (R) of a power series $\sum_{n=0}^{\infty} a_n(x-c)^n$ is defined to be number
  - $R = 0$ if the series is divergent for all $x \neq c$.
  - $R = \infty$ if the series is absolutely convergent for all x.
  - R, the positive member such that the series converges absolutely for all real x satisfying $|x-c| < R$ and diverges for all x satisfying $|x-c| > R$.

Computation of Radius of Convergence

- Ratio Test for power series: Consider the power series $\sum_{n=0}^{\infty} a_n(x-c)^n$. Then the radius of convergence is given as follows:

$$\frac{1}{R} = \lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right|$$

- Theorem: Root Test for power series: Consider the power series $\sum_{n=0}^{\infty} a_n(x-c)^n$. Then the radius of convergence is given as follows:

$$\frac{1}{R} = \limsup_{n\to\infty} \sqrt[n]{|a_n|}$$

Finding Interval/Region of Convergence

- If, $\frac{1}{R} = 0 \implies R = \infty \implies$ the series is absolutely convergent for all x.
- If, $\frac{1}{R} = \infty \implies R = 0 \implies$ the series is divergent for all $x \neq c$.
- If, $\frac{1}{R} = A$ (finite number) $\implies R = \frac{1}{A} \implies$ the series converges absolutely for all real x satisfying $|x - c| < R = \frac{1}{A}$ and diverges for all x satisfying $|x - c| > R = \frac{1}{A}$.

  Note: On the boundary points, the series may or may not be convergent. So, one has to check manually whether the power series is convergent at the boundary points or not.

Power series : Examples

- The following power series is centered at 0.

$$\sum_{n=0}^{\infty} x^n$$

This is the geometric series. It converges for $|x| < 1$ and diverges for $|x| \geq 1$.

- The following power series is centered at 0.

$$\sum_{n=0}^{\infty} \frac{x^n}{n!}$$

This series converges absolutely for all x. (using Ratio test)

- The power series $\displaystyle\sum_{n=0}^{\infty} (x-1)^n$ is centered at 1. It converges for $|x-1| < 1$ and diverges for $|x-1| \geq 1$.

More Examples

- Find the radius of convergence and interval of Convergence of

(i) $\sum \frac{x^n}{n}$, (ii) $\sum \frac{x^n}{n!}$, (iii) $\sum 2^{-n}x^n$.

(i) $\frac{1}{R} = \lim_{n\to\infty} |\frac{a_{n+1}}{a_n}| = 1,$. So $R = 1$ and the series converges absolutely for all real x satisfying $|x| < R = 1$ and diverges for all x satisfying $|x| > R = 1$.

(ii) $\frac{1}{R} = \lim_{n\to\infty} |\frac{a_{n+1}}{a_n}| = 0$. So $R = \infty$, and series converges everywhere.

(iii) $\frac{1}{R} = \lim_{n\to\infty} \sqrt[n]{|a_n|} = 2^{-1} = \frac{1}{2}$. Therefore, $R = 2$ and the series converges absolutely for all real x satisfying $|x| < R = 2$ and diverges for all x satisfying $|x| > R = 2$.

Taylor's Series

Brook Taylor (1685–1731)

Taylor series

Taylor's series

The Taylor series of a real-valued function $f(x)$ that is infinitely differentiable at a real number c is the power series

$$f(x) = f(c) + \frac{f'(c)}{1!}(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \frac{f'''(c)}{3!}(x-c)^3 + ...$$

$$\implies f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!}(x-c)^n.$$

where $f^{(n)}(c)$ denotes the nth derivative of f evaluated at the point c.

Maclaurin's Series

Maclaurin's series

- If $c = 0$, the formula obtained in Taylor's theorem is known as *Maclaurin's series*

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \ldots$$

$$\Longrightarrow f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n.$$

Here $f^{(n)}(0)$ represents n-th derivative of the function $f(x)$ evaluated at $x = 0$.

Examples

Examples

(i) Find Taylor series of $f(x) = e^x$ about $c = 0$.

We have $f^{(n)}(x) = e^x$. So $f^{(n)}(0) = e^0 = 1$.

$$f(x) = e^x = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}(x-0)^n = \sum_{n=0}^{\infty} \frac{x^n}{n!}.$$

(ii) Find Taylor series of $f(x) = e^x$ about $c = -1.5$.

We have $f^{(n)}(x) = e^x$. So $f^{(n)}(-1.5) = e^{-1.5}$.

$$f(x) = e^x = \sum_{n=0}^{\infty} \frac{f^{(n)}(-1.5)}{n!}(x+1.5)^n = \sum_{n=0}^{\infty} \frac{e^{-1.5}}{n!}(x+1.5)^n.$$

Riemann Integration

Bernhard Riemann (1826–1866)

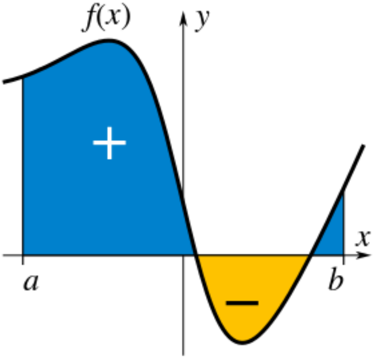
BENNETT
UNIVERSITY
TIMES OF INDIA GROUP

Integration

- An **integral** assigns numbers to functions in a way that describes displacement, area, volume, and other concepts that arise by combining infinitesimal data.
- The process of finding integrals is called **integration**.
- Integrals can be categorized into two types.
  1. Definite integrals
  2. Indefinite integrals

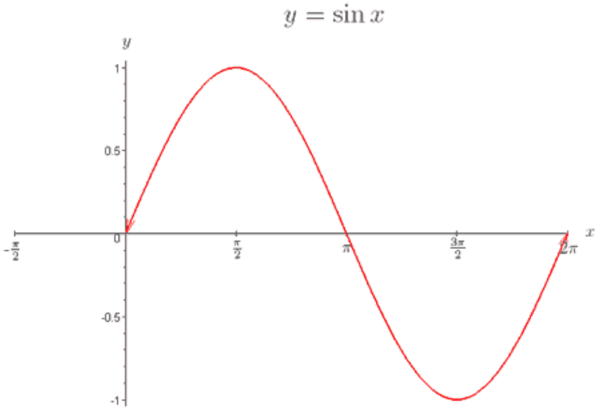
Definite Integrals

- The definite integrals can be interpreted as the signed area of the region in the plane that is bounded by the graph of a given function between two points in the real line.
- Conventionally, areas above the horizontal axis of the plane are positive while areas below are negative.



- **Example :**

$$\int_0^{2\pi} \sin x \; dx = \Big[-\cos x \Big]_0^{2\pi} = 0$$



Benefits/Use of Integrals

- Integrals appear in many practical situations. For instance, from the length, width and depth of a swimming pool which is rectangular with a flat bottom, one can determine the volume of water it can contain, the area of its surface, and the length of its edge.
- But if it is oval with a rounded bottom, integrals are required to find exact and rigorous values for these quantities. In each case, one may divide the sought quantity into infinitely many infinitesimal pieces, then sum the pieces to achieve an accurate approximation.

Interpretations of Integrals

- **Example :** To find the area of the region bounded by the graph of the function $f(x) = \sqrt{x}$ between $x = 0$ and $x = 1$, one can cross the interval in five steps $(0, 1/5, 2/5, ..., 1)$, then fill a rectangle using the right end height of each piece (thus $\sqrt{0}, \sqrt{1/5}, \sqrt{2/5}, ..., \sqrt{1}$) and sum their areas to get an approximation of

$$\sqrt{\frac{1}{5}}\left(\frac{1}{5} - 0\right) + \sqrt{\frac{2}{5}}\left(\frac{2}{5} - \frac{1}{5}\right) + \cdots + \sqrt{\frac{5}{5}}\left(\frac{5}{5} - \frac{4}{5}\right) \approx 0.7497,$$

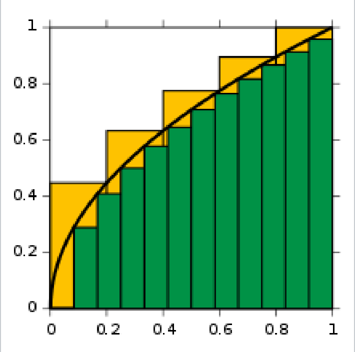
  which is larger than the exact value.

- Alternatively, when replacing these subintervals by ones with the left end height of each piece, the approximation one gets is too low: with twelve such subintervals the approximated area is only 0.6203.

- However, when the number of pieces increase to infinity, it will reach a limit which is the exact value of the area sought (in this case, $2/3$). One writes

$$\int_0^1 \sqrt{x}\,dx = \frac{2}{3},$$

which means $2/3$ is the result of a weighted sum of function values, $\sqrt{x}$, multiplied by the infinitesimal step widths, denoted by dx, on the interval $[0, 1]$.



Approximations to integral of $\sqrt{x}$ from 0 to 1, with 5 yellow right endpoint partitions and 12 green left endpoint partitions

Some links for further Visualization

For upper sum of the function $y = x^2$, follow the below link.

https://en.wikipedia.org/wiki/File:Riemann_Integration_and_Darboux_Upper_Sums.gif

For lower sum

https://en.wikipedia.org/wiki/File:Riemann_Integration_and_Darboux_Lower_Sums.gif

Riemann Integrals

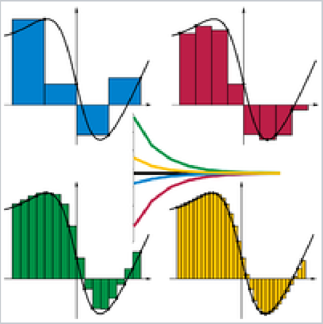
- The Riemann integral is defined in terms of Riemann sums of functions with respect to **tagged partitions** of an interval.
- A tagged partition of a closed interval $[a, b]$ on the real line is a finite sequence

$$a = x_0 \leq t_1 \leq x_1 \leq t_2 \leq x_2 \leq \cdots \leq x_{n-1} \leq t_n \leq x_n = b.$$

- This partitions the interval $[a, b]$ into n sub-intervals $[x_{i-1}, x_i]$ indexed by i, each of which is "tagged" with a distinguished point $t_i \in [x_{i-1}, x_i]$.
- A Riemann sum of a function f with respect to such a tagged partition is defined as

$$\sum_{i=1}^{n} f(t_i)\,\Delta_i.$$

- Each term of the sum is the area of a rectangle with height equal to the function value at the distinguished point of the given sub-interval, and width the same as the width of sub-interval, $\Delta_i = x_i - x_{i-1}$.



Riemann sums converging

- The Riemann integral of a function f over the interval $[a, b]$ is equal to S if

For all $\varepsilon > 0$, there exists $\delta > 0$ such that, for any tagged partition $[a, b]$ with mesh less than δ,

$$\left| S - \sum_{i=1}^{n} f(t_i)\,\Delta_i \right| < \varepsilon.$$

Riemann Integrals (simplified)

To visualize a sequence of Riemann sums over a regular partition of an interval, follow the below link.

https://upload.wikimedia.org/wikipedia/commons/2/28/Riemann_integral_regular.gif

Simplified formulation of Riemann integral:

$$\boxed{\int_a^b f(x)dx = \lim_{\|\Delta x\|\to 0} \sum_{k=1}^{n} f(t_k)(x_k - x_{k-1}).}$$

Some other integrals (not in our course, but for more curious students)

1. Cauchy integral
2. Riemann-Stieltjes integral
3. Lebesgue integral
4. Lebesgue-Stieltjes integral
5. Daniell integral
6. Haar integral
7. Henstock-Kurzweil (HK) integral
8. Wiener integral
9. Feynman integral

Question

How can we calculate the value of a definite integral?

Is it always by using the definition, i.e. by the limit-of-sum approach?

Or, is there any easy approach?

Question

How can we calculate the value of a definite integral?

Is it always by using the definition, i.e. by the limit-of-sum approach?

Or, is there any easy approach?

Rescue: Fundamental Theorem of Calculus

Fundamental Theorem of Calculus

- The fundamental theorem of calculus is a theorem that links the concept of differentiating a function (calculating the gradient) with the concept of integrating a function (calculating the area under the curve).
- The two operations are inverses of each other apart from a constant value which depends where one starts to compute area
- So, we call integrals as anti-derivatives.

Fundamental Theorem of Calculus

Let f be a continuous real-valued function defined on a closed interval $[a, b]$.

Let F be the function defined, for all x in $[a, b]$, by

$$F(x) = \int_a^x f(t)\,dt.$$

Then F is uniformly continuous on $[a, b]$ and differentiable on the open interval (a, b), and

$$F'(x) = f(x) \text{ for all } x \in (a, b).$$

Application of Fundamental Theorem of Calculus

- The fundamental theorem is often employed to compute the definite integral of a function f for which an anti-derivative F is known.
- Let f be a real-valued function on a closed interval $[a, b]$ and F an anti-derivative of f in (a, b) such that

$$F'(x) = f(x).$$

  If f is Riemann integrable on $[a, b]$, then

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

- **Example :** Suppose the following is to be calculated:

$$\int_2^5 x^2\,dx.$$

Here, $f(x) = x^2$ and we can use $F(x) = \dfrac{x^3}{3}$ as the anti-derivative.

$$\therefore \quad \int_2^5 x^2\,dx = F(5) - F(2) = \frac{5^3}{3} - \frac{2^3}{3} = \frac{125}{3} - \frac{8}{3} = \frac{117}{3} = 39.$$

Properties of Definite Integrals

- Linearity: $\int_a^b (\alpha f + \beta g)(x)\, dx = \alpha \int_a^b f(x)\, dx + \beta \int_a^b g(x)\, dx.$
- $\int_a^b f(x)\, dx = -\int_b^a f(x)\, dx.$
- $\int_a^b f(x)\, dx = \int_a^c f(x)\, dx + \int_c^b f(x)\, dx.$
- An integrable function f on $[a, b]$, is necessarily bounded on that interval. Thus there are real numbers m and M so that $m \leq f(x) \leq M$ for all $x \in [a, b]$. Hence we have

$$m(b-a) \leq \int_a^b f(x)\, dx \leq M(b-a).$$

Area Between Curves

Area Between Curves

Let $f(x)$ and $g(x)$ be continuous functions defined on $[a, b]$ where $f(x) \geq g(x)$ for all $x \in [a, b]$. The area of the region R bounded by the curves $y = f(x), y = g(x)$ and the lines $x = a$ and $x = b$ is

$$\int_a^b \Big(f(x) - g(x)\Big) dx.$$

Examples

- Find the area of the region enclosed by $y = x^2$ and $y = 2x + 3$.

  Ans: These two curves intersect at $x = -1$ or $x = 3$.

$$
\begin{aligned}
\therefore \quad \int_{-1}^{3} \left(2x + 3 - x^2\right) dx &= \left(x^2 + 3x - \frac{x^3}{3}\right)\Bigg|_{-1}^{3} \\
&= (9 + 9 - 9) - (1 - 3 + \frac{1}{3}) \\
&= 9 + \frac{5}{3} \\
&= \frac{32}{3} \text{ sq. units}
\end{aligned}
$$

- Find the area of a circle.

  Ans: First we'll calculate the area of a quarter (preferable which lie in the first quadrant). Then we'll multiply the result by 4 to get the final result.

  $\therefore$ The area of a circle $x^2 + y^2 = r^2$ is given by

$$4\int_0^r \sqrt{r^2 - x^2}dx = 4\left(\frac{x}{2}\sqrt{r^2 - x^2} + \frac{r^2}{2}\sin^{-1}\frac{x}{r}\right)\Bigg|_0^r$$
$$= \frac{r^2}{2}\sin^{-1} 1 = 4 \times \pi\frac{r^2}{4}$$
$$= \pi r^2 \text{ sq. units}$$

- Find the area of the region enclosed by the functions $f(x) = \sqrt{x} + 2, g(x) = -(x-1)^2 + 3$ and $h(x) = 2$.

  Ans: $\frac{4}{3}$ (Exercise)

Appendix

Some Useful Formulae

- $\displaystyle\int x^n dx = \frac{x^{n+1}}{n+1} + C; \quad n \neq 1$
- $\displaystyle\int \sin(ax+b)\ dx = -\frac{1}{a}\cos(ax+b) + C, \quad a \neq 0$
- $\displaystyle\int \cos(ax+b)\ dx = \frac{1}{a}\sin(ax+b) + C, \quad a \neq 0$
- $\displaystyle\int \sec^2 x\ dx = \tan x + C$
- $\displaystyle\int \csc^2 x\ dx = -\cot x + C$
- $\displaystyle\int \sec x \cdot \tan x\ dx = \sec x + C$
- $\displaystyle\int \csc x \cdot \cot x\ dx = -\cot x + C$
- $\displaystyle\int \frac{dx}{x} = \ln|x| + C$

- $\int e^{ax+b} dx = \frac{e^{ax+b}}{a}, \quad a \neq 0$
- $\int a^x dx = \frac{a^x}{\ln a} + C, \quad a \neq 0$
- $\int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1} x + C$
- $\int \frac{1}{1+x^2} dx = \tan^{-1} x + C$
- $\int \frac{1}{|x|\sqrt{x^2-1}} dx = \sec^{-1} x + C$
- $\int \sqrt{x^2-a^2} dx = \frac{x}{2}\sqrt{x^2-a^2} - \frac{a^2}{2} \ln |x + \sqrt{x^2-a^2}| + C$
- $\int \sqrt{x^2+a^2} dx = \frac{x}{2}\sqrt{x^2+a^2} + \frac{a^2}{2} \ln |x + \sqrt{x^2+a^2}| + C$
- $\int \sqrt{a^2-x^2} dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} + C$

THANK YOU.